Automated Deployment of Genome Databases for Marine Brown Algae using Galaxy Genome Annotation

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Genomic context: the PhaeoExplorer project

Data:

- Genome and transcriptome data for 40 brown algae and 4 sister species
- A total of 62 strains (M & F), 20 high quality genomes and 42 drafts
- Complete annotated genomes based on Illumina PE DNA-seq and RNA-seq plus Oxford Nanopore or 10x Genomics data for a subset of genomes

Goal:

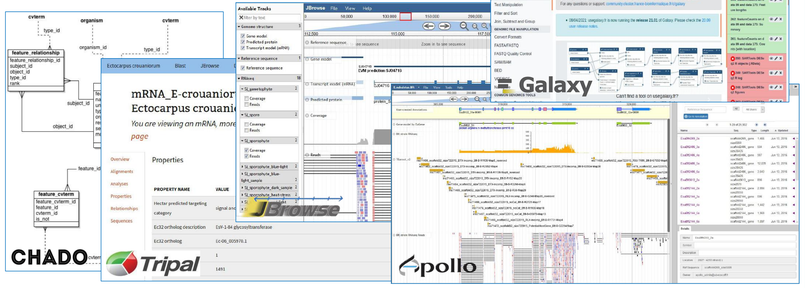
Establish a genomic data resource designed specifically to facilitate investigation of the emergence and diversification of key biological traits within the brown algae, particularly those associated with the transition to complex multicellularity.

How to store ? Visualize ? Explore ? Analyze ? Edit ? Share ? Publish ?

GMOD tools

A set of interconnected open source applications and databases to visualize, annotate and manage genomic data.

An active community.



Manual deployment of all applications is not reasonable as the number of genomes resources to integrate is growing.

Galaxy Genome Annotation project

Community

Initiative from the Galaxy and the GMOD communities
doi: 10.7490/f1000research.1116992.1
Open-source licenses

A. Bretaudeau, H. Rasche, N. Dunn

Facilitating genome annotation using Galaxy and GMOD tools

https://galaxy-genome-annotation.github.io

Galaxy tools

Perform structural or functional annotation

Gene prediction: **Prokka, Maker, Augustus, SNAP**
Functional annotation : **Prokka, Antismash**
Quality assessment: **BUSCO**
Repeated elements: **RepeatMasker**
Visualisation: **JBrowse, Circos**

GMOD integration

Docker images for GMOD tools
Python libraries to make them work together
Galaxy tools to interact with GMOD applications

Galaxy flavor

https://annotation.usegalaxy.eu/

Training materials

Deployment of GMOD tools with Galaxy Genome Annotation

Implementation:

1 genome = 1 set of reusable dockerized web applications
Galaxy as data loading orchestrator for administrators: tools and workflows to load data into JBrowse, Apollo, Tripal, Chado

Benefits:

Automatic deployment : Human intervention reduced to the minimum
Reproducibility: 1 genome = docker-compose.yml + raw data (fasta, gff, bam) + Galaxy workflow
Flexibility : Customizable set of applications for each genome, all configured in a single file
Scalability: Possibility to dispatch Docker containers on a cluster of Docker hosts (e.g. docker-swarm)

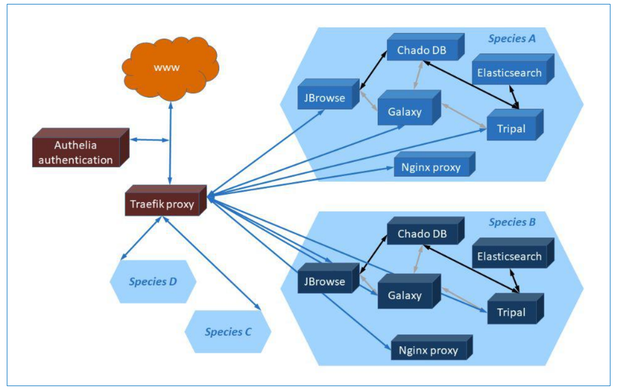


*Figure : **Schematic representation of a set of Docker containers deployed with GGA for typical genomes.** Cuboids represent Docker containers. Hexagons represent different sets of Docker containers. Blue arrows represent HTTP traffic. Gray arrows represent data exchange performed using Galaxy tools. Black arrows represent data exchange inherent in applications.*

Fully automated deployment of GMOD visualisation applications for a set of genomes and datasets

A pipeline of Python tools allowing mass deployment of above GGA services and automated data loading through Galaxy has been developed using Galaxy Bioblend API and Galaxy workflows:

http://gitlab.sb-roscoff.fr/abims/e-infra/gga_load_data **Release 2.1.0**

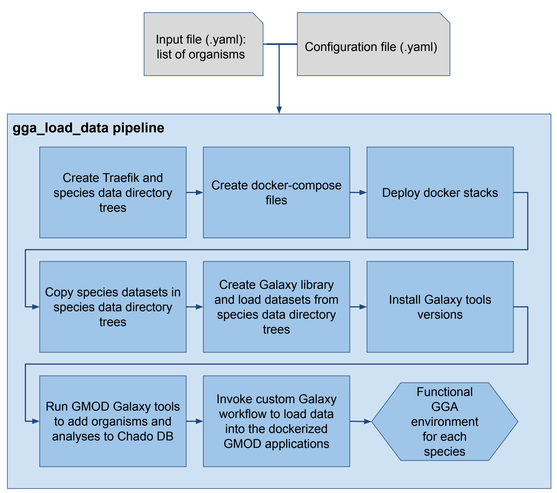


*Figure : **Flow chart of the gga_load_data pipeline***

The goal of the pipeline is to deploy a GGA environment for a batch of organisms without the need for manual configuration and installation.

Each step is performed with different Python scripts that interact with Docker and the Galaxy instance deployed for each input species.

The gga_load_data pipeline takes care of initializing the Docker swarm and deploying/updating the different docker stacks used by GGA.

*Galaxy interaction is handled using the **Bioblend** Python API to remotely perform actions in the different Galaxy instances programmatically.*

The workflow invocation script is designed to work with PhaeoExplorer data, as some GMOD Galaxy tools require some parameters to be adapted to the dataset format and nomenclature, although this script can be modified to support different datasets.

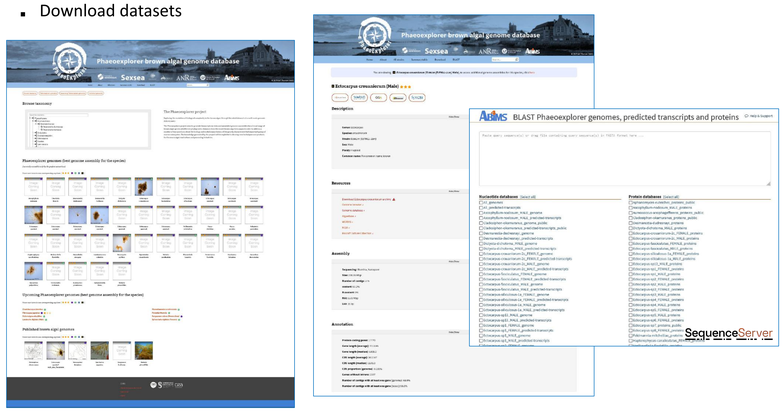
PhaeoExplorer web portal

A hub for accessing, visualizing and analyzing the brown algal genome and transcriptome resources

https://phaeoexplorer.sb-roscoff.fr

Available resources:

- a GGA tool suite for each species:
  - Tripal as GGA entry point, with a search field
  - a Tripal page for each gene with analysis results (*E. species Ec32* ortholog, HECTAR, …)
  - a genome browser (JBrowse) to visualize gene models mapped on the genome
- BLAST interfaces: set up with SequenceServer (https://sequenceserver.com/) and Ansible (https://galaxy.ansible.com/abims_sbr/sequenceserver)
- Metadata and annotation statistics
- Links to external databases (NCBI, WoRMS, Algaebase, …)
- Download datasets



Future developments

Upcoming PhaeoExplorer developments:

- Deploy GGA services for public genomes: *N. decipiens, S. fusiforme, U. pinnatifida, C. okamuranus, E. species Ec32*
- Add functional annotations (InterproScan, Blast vs uniref90)
- Add RNAseq data
- Add orthology module (GGA environment to visualize orthologs between species)

Other developments:

- Ongoing: cloudify the GGA tool suite and implement in the European Open Science Cloud (see poster #126)
- Deploy a similar web portal for red algal genomes and for other marine transcriptomic data
- Support for more data types (expression, genotype, phenotype, etc.)
- Links to genome-scale metabolic networks (http://aureme.genouest.org)
- Provide these collaborative GGA environments to other scientific consortiums (ERGA, VGP)

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